



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course Fundamentals of Data Science

Code DSA0404

Slot A

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Unit 4 - Statistical Inference

CO4-AT3 - Viva Voce Questions



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A. INTRODUCTION TO STATISTICAL INFERENCE

1. What is the statistical inference?

Statistical inference is the process of using sample data to draw conclusions about a larger population.

2. Why is statistical inference important in data science?

Statistical inference helps data scientists make decisions, estimate unknown population values, and test claims using sample data.

3. What is the difference between descriptive statistics and inferential statistics?

Descriptive statistics summarize the data that has been collected, while inferential statistics uses sample data to make conclusions about a larger population.

4. What is a population in statistics?

A population is the complete set of individuals, objects, or observations that we are interested in studying.

5. What is a sample?

A sample is a smaller group of observations selected from the population for analysis.

6. Why do data scientists usually work with samples instead of complete populations?

Data scientists usually work with samples because studying the entire population can be expensive, time-consuming, or practically impossible.

7. What is a population parameter?

A population parameter is a numerical value that describes a characteristic of the entire population. Examples include population mean, population proportion, and population variance.

8. What is a sample statistic?

A sample statistic is a numerical value calculated from sample data. Examples include sample mean, sample proportion, and sample variance.

9. Give a real-world example where statistical inference is used in industry.

A food delivery company can select 100 deliveries and use their average delivery time to estimate the average delivery time for all deliveries.

10. What are the two major activities involved in statistical inference?

The two major activities are estimation and hypothesis testing.

B. FREQUENTIST APPROACH

11. What is the frequentist approach to statistical inference?

The frequentist approach uses sample data and long-run probabilities to make conclusions about unknown population parameters.

12. What is the basic idea behind frequentist statistics?

The basic idea is that probability describes the long-run frequency of events when an experiment is repeated many times under similar conditions.

13. How does the frequentist approach differ from the Bayesian approach?

The frequentist approach treats the population parameter as fixed and the data as random. The Bayesian approach represents uncertainty about the parameter using a probability distribution.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

The population parameter is considered fixed, while the sample data is considered random.

15. What does long-run frequency mean in frequentist statistics?

Long-run frequency means the proportion of times an event occurs when an experiment is repeated many times under the same conditions.

16. Why is random sampling important in the frequentist approach?

Random sampling helps ensure that the sample is representative of the population and supports valid statistical inference.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

No. In the frequentist approach, a population parameter is considered a fixed but unknown value.

18. Give an example of a frequentist problem in a data science application.

A company may test whether a new recommendation algorithm significantly increases the conversion rate compared with an existing algorithm.

C. POINT ESTIMATION

19. What is a point estimate?

A point estimate is a single numerical value used to estimate an unknown population parameter.

For example, if the sample mean is 32 minutes, then 32 minutes is the point estimate of the population mean.

20. What is an estimator?

An estimator is a rule or formula used to calculate an estimate of a population parameter from sample data.

21. What is the difference between an estimator and an estimate?

An estimator is the formula or rule used for estimation, whereas an estimate is the actual numerical value obtained from the sample.

For example:

Estimator: $\bar{x} = \sum x / n$

Estimate: $\bar{x} = 32$ minutes

22. What is the point estimator for the population mean?

The sample mean, represented by $\bar{x}$, is the point estimator for the population mean.

Formula:

$$\bar{x} = \sum x / n$$

23. How do you estimate a population proportion from a sample?

A population proportion is estimated using the sample proportion.

Formula:

$$\hat{p} = x / n$$

where x is the number of successes and n is the sample size.

24. What is the point estimate of population variance?

The sample variance, represented by s^2 , is used as the point estimator of population variance.

Formula:

$$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$$

25. Why is the sample mean commonly used to estimate the population mean?

The sample mean is commonly used because it is an unbiased estimator of the population mean and becomes more reliable as the sample size increases.

26. What properties should a good estimator have?

A good estimator should have important properties such as unbiasedness, consistency, and efficiency.

27. What is an unbiased estimator?

An estimator is unbiased if its expected value is equal to the true population parameter.

For the sample mean:

$$E(\bar{x}) = \mu$$

28. What is meant by the bias of an estimator?

Bias is the difference between the expected value of an estimator and the true population parameter.

Formula:

$\text{Bias} = E(\text{Estimator}) - \text{Parameter}$

29. What is the difference between bias and variance of an estimator?

Bias measures the systematic difference between an estimator and the true parameter, while variance measures how much the estimator changes from one sample to another.

30. Why can two different samples produce different point estimates for the same population parameter?

Two different random samples can contain different observations. Therefore, their sample statistics, such as sample means, can also be different.

D. VARIABILITY IN ESTIMATES

31. Why does a point estimate have uncertainty?

A point estimate has uncertainty because it is calculated from a sample rather than the entire population. Another sample may produce a different estimate.

32. What is the sampling distribution of an estimator?

The sampling distribution is the probability distribution of an estimator obtained from repeated samples taken from the same population.

33. What is the standard error?

Standard error measures the variability of a sample statistic across repeated samples.

For the sample mean:

$$SE = \sigma / \sqrt{n}$$

When the population standard deviation is unknown:

$$SE = s / \sqrt{n}$$

34. How is standard error different from standard deviation?

Standard deviation measures the variability among individual observations, while standard error measures the variability of a sample statistic such as the sample mean.

35. How does increasing the sample size affect the standard error of the sample mean?

Increasing the sample size decreases the standard error.

Formula:

$$SE = \sigma / \sqrt{n}$$

As n increases, SE decreases.

36. Why does variability in an estimate decrease when the sample size increases?

A larger sample contains more information about the population, making the estimate more stable and less affected by random sampling variation.

37. What is the relationship between standard error and confidence intervals?

Standard error determines the margin of error and therefore affects the width of a confidence interval.

Formula:

$$\text{Confidence Interval} = \text{Estimate} \pm \text{Critical Value} \times \text{SE}$$

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.

No. Different random samples can naturally produce different sample means because of sampling variability. Different estimates do not necessarily mean that either sample is incorrect.

E. CONFIDENCE INTERVALS

39. What is a confidence interval?

A confidence interval is a range of values used to estimate an unknown population parameter with a specified confidence level.

For example:

95% CI = (30, 35)

40. Why is a confidence interval more informative than a point estimate?

A point estimate gives only one value, while a confidence interval provides a range of plausible values and shows the uncertainty associated with the estimate.

41. What does a 95% confidence level mean in the frequentist approach?

If we repeatedly take samples and construct 95% confidence intervals using the same method, approximately 95% of those intervals would contain the true population parameter.

42. What is the difference between a confidence level and a confidence interval?

A confidence level is the percentage used to describe the reliability of the estimation method, such as 95%. A confidence interval is the actual numerical range calculated from the sample.

43. What factors determine the width of a confidence interval?

The main factors are sample size, variability or standard error, and the selected confidence level.

44. How does increasing the sample size affect the width of a confidence interval?

Increasing the sample size generally makes the confidence interval narrower because the standard error decreases.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

Increasing the confidence level from 95% to 99% makes the confidence interval wider because a larger critical value is required.

46. What is the difference between a z-confidence interval and a t-confidence interval?

A z-confidence interval is generally used when the population standard deviation is known or when normal approximation conditions are appropriate. A t-confidence interval is used when the population standard deviation is unknown and is estimated using the sample standard deviation.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

The t-distribution is used when the population standard deviation is unknown and the sample standard deviation is used, particularly when the sample size is small.

F. HYPOTHESIS TESTING USING CONFIDENCE INTERVALS

48. What is hypothesis testing?

Hypothesis testing is a statistical method used to determine whether there is sufficient evidence to support or reject a claim about a population.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

The null hypothesis, H_0 , represents the initial claim or no significant effect or difference.

The alternative hypothesis, H_1 , represents the claim that there is a significant effect or difference.

Example:

$$H_0: \mu = 30$$

$$H_1: \mu \neq 30$$

50. How can a confidence interval be used to make a decision about a hypothesis?

The claimed population parameter is compared with the confidence interval.

For example:

$$H_0: \mu = 30$$

$$95\% \text{ CI} = (31.5, 35.2)$$

Since 30 is outside the confidence interval, the null hypothesis is rejected at the corresponding significance level.

IMPORTANT FORMULAS FOR VIVA

Sample Mean

$$\bar{x} = \Sigma x / n$$

Sample Proportion

$$\hat{p} = x / n$$

Sample Variance

$$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$$

Standard Error of Mean

$$SE = s / \sqrt{n}$$

Confidence Interval

$$CI = \text{Estimate} \pm \text{Critical Value} \times SE$$

Bias

$$\text{Bias} = E(\text{Estimator}) - \text{Parameter}$$

Main Activities of Statistical Inference

Estimation + Hypothesis Testing

Frequentist Approach

Parameter = Fixed

Data = Random